

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Third Semester of BSc Examination April 2018
BS307 Basic Biostatistics

Date: 07/04/2018, Saturday

Time: 01:30 p.m. to 01:40 p.m.

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer/s and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

1. A statistician records the height of school children of age 10-12 years, hence data on height is of type
 - (i) continuous
 - (ii) discrete
 - (iii) nominal
 - (iv) ordinal
2. The arithmetic mean of a series of variable X_1, X_2, X_3, X_4, X_5 is _____.
 - (i) $X_1 + X_2 + X_3 + X_4 + X_5$
 - (ii) $\frac{5}{\frac{1}{X_1} + \frac{1}{X_2} + \frac{1}{X_3} + \frac{1}{X_4} + \frac{1}{X_5}}$
 - (iii) $(X_1 \times X_2 \times X_3 \times X_4 \times X_5)^{\frac{1}{5}}$
 - (iv) $\frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}$
3. The Arithmetic mean of six observations is 7. A new observation 7 is added. The mean of seven observations is _____.
 - (i) 14
 - (ii) 8.16
 - (iii) 7
 - (iv) 2

4. The Median of the variate values 11,7,6,9,12,15,19 is_____.
- (i) 9 (iii) 15
(ii) 12 (iv) 11
5. If the variance of a distribution is 9, the standard deviation of distribution is _____.
- (i) 3 (iii) 9
(ii) 6 (iv) 81
6. The standard deviation of a group of scores is 10. If 5 were subtracted from each score, the standard deviation of the new scores would be _____.
- (i) 2 (iii) 10/25
(ii) 10 (iv) 5
7. A researcher obtained the regression line $\hat{Y} = -3.6 + 3.1X$, where X is diameter of tree trunk and Y is tree height. The estimate of the average heights of all trees having a trunk diameter of 7 inches is _____.
- (i) 6.5 (iii) 18.1
(ii) 25.3 (iv) 21.7
8. The simple correlation coefficient between variables X and Y is known to be zero. We then can conclude that _____.
- (i) the variances of X and Y are equal (iii) there exists no linear relationship between X and Y
(ii) the covariance between X and Y is zero (iv) Both (ii) and (iii)
9. If b_{yx} and b_{xy} respectively denote the linear regression coefficients of regression line Y on X and regression line X on Y , then the linear correlation coefficient $r_{xy} =$ _____.
- (i) $b_{yx} \times b_{xy}$ (iii) $b_{yx} + b_{xy}$
(ii) $\pm \sqrt{b_{yx} \times b_{xy}}$ (iv) $\pm \sqrt{b_{yx} + b_{xy}}$
10. Given the two lines of regression as, $3X - 4Y + 8 = 0$ and $4X - 3Y = 1$, the arithmetic mean $\bar{X} =$ _____ and $\bar{Y} =$ _____.
- (i) 4,5 (iii) $\frac{4}{3}, \frac{5}{4}$
(ii) 3,4 (iv) -4,-5

End of MCQ

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Date: 07/04/2018, Saturday

Time: 01:40 p.m. to 03:30 p.m.

Maximum Marks: 25

Instructions:

- Section I and II must be attempted in SINGLE ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION I

Q - II Answer the following questions as directed. [10]

1. Define the following terms. [04]
 - (a) Mean and Median
 - (b) Mathematical Properties of arithmetic mean.
2. The following table shows the frequency distribution of live births according to age of mother.

Table 1: Live Births and Mother's Age

Mother's Age	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49
Number of Live Births	7	258	585	841	981	526	99	4

From the frequency distribution in Table 1, Compute [06]
 (i) Arithmetic Mean Age (ii) Median Age.

SECTION II

Q - III Answer the following questions as directed (Attempt any three). [15]

1. The Table 2 below shows prolactin levels (ng/L) obtained during a clinical trial involving 10 subjects.

Table 2: prolactin levels (ng/L)

9.4	7.0	7.6	6.3	6.7
8.6	6.8	10.6	8.9	9.4

Compute for prolactin level (i) Arithmetic Mean (ii) Standard deviation. [05]

2. A Clinical Trial conducted to determine the pharmacokinetic parameters for a new calcium channel blocker. The Table 3 below shows maximum serum concentration (ng/ml) of 10 subjects.

Table 3: $C_{\max}$ data ng/ml

715	728	735	716	706	715	712	717	731	709
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Compute (i) All Quartiles (ii) Mean deviation about Median.

[05]

3. It is believed that two assay methods will produce identical results for analyzing a specific drug (Table 4). Various dilutions are assayed using both the currently accepted method (GS) and the proposed alternative (ALT).

Table 4: results of two assay methods

Method GS	90.1	85.2	79.7	74.3	60.2	35.5	24.9	19.6
Method ALT	89.8	85.1	80.2	75.0	61.0	34.8	24.8	21.1

Compute the coefficient of simple correlation between the results of two assay methods. Write your conclusion from your computations.

[05]

4. During the formulation of a new product, various percent of a polymeric coating are added and the resultant release of the therapeutic agent is evaluated below. (Table 5)

Table 5: Data for regression

Percent Coating	5	10	15	20	30	40
Rate of Release (mg/hr)	3.70	3.00	2.74	2.34	1.41	0.56

Obtain the least squares line to predict the rate of release of therapeutic agent.

[05]